Functional Description of Diagnosis on

CAN

For

ABS10.3MB TVS U712_13

System: **Customer:** **Vehicle:**	Front/Rear Brake Split. 2 Channel ABS10.3 MB System	**Equipment:** **Tester/Diagnosis:** CAN
Distribution:	Customer, Internal	

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Please refer signature workflow

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History

Version No.	Date	Editor	Reviewer	Description of changes
1.0	23-01-2024	Gautham N S	Naik Vinuta Vittala	• Based on 0265T82041e_4e • Updated Section 6,12,13 and 19 DTC(Drag Torque control)

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1 Referenced Documents

ISO documents

The general requirements of diagnosis communication described in this document are based on the Unified diagnostic services (UDS) design standards published by ISO (International Organization for Standardization) with the restrictions / additions as defined by the subsequent sub-sections:

ISO 14229: Road Vehicles – Diagnostic Systems: Diagnostic Services Specification
ISO 15765: Road Vehicles- Diagnostic communication over Controller Area Network

2 Introduction

UDS based CAN diagnosis provides the capability of quickly accessing the information and functions of the ECU over the CAN, without disconnecting the cable harness. This diagnosis provides services to:

- Identify the ECU.
- Read input signals applied to the ECU.
- Actuate all switching elements connected to the ECU.
- Read out internal information, such as the contents of the fault memory etc., stored in the ECU.
- Execute ROM resident test sequence routines.

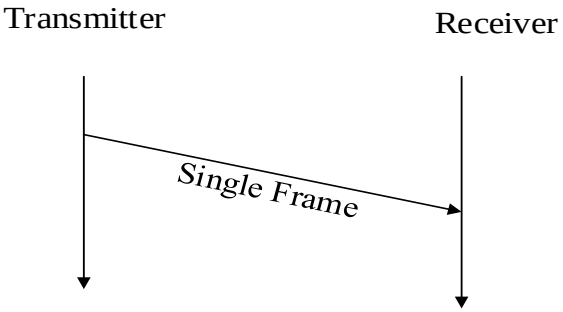
3 Communication via CAN

3.1 Data stream

The ECU supports CAN diagnostic messages using the Unacknowledged Segmented Data Transfer (USDT) mechanism as specified in ISO 15765-2. Accordingly, the following two types of CAN data streams are supported:

3.1.1 Non-segmented data stream

Consists of a Single Frame (SF) of non-segmented CAN message. This shall be used for messages with maximum data length of 7-Bytes.



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3.1.2 Segmented data stream

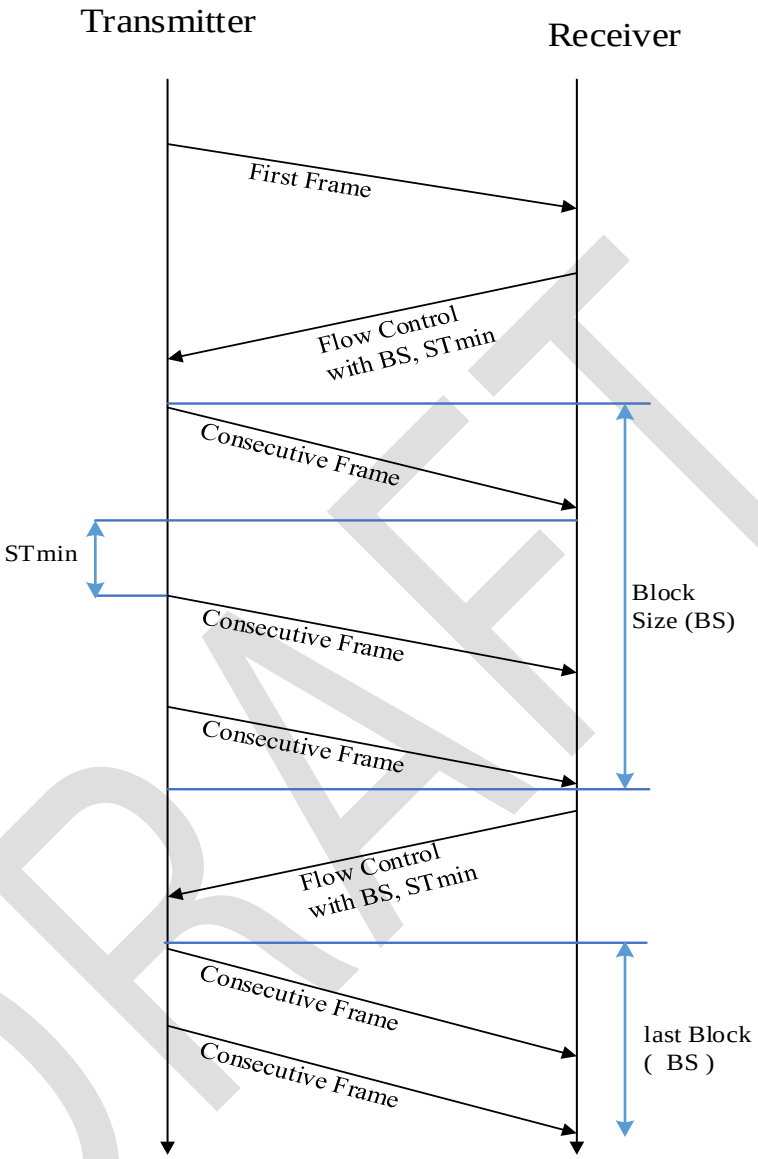
Consists of multiple frames of segmented CAN messages. This shall be used when the maximum data length exceeds 7-Bytes.

The data stream comprises of a First Frame (FF) and several Consecutive Frames (CF) containing the subsequent data bytes that follow the First Frame. A Handshake, in the form of a Flow Control (FC), shall be send by the receiver node after reception of the First Frame.

The Flow Control shall contain information regarding the maximum number of Consecutive Frames the receiver shall allow (Block Size –BS) after which the transmitter shall wait for an authorisation from the receiver to continue transmission, and the minimum time the transmitter shall wait between transmission of two consecutive frames (Separation Time -STmin).

Note that functional addressing does not support segmented data streams.

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Transmitter
Receiver
First Frame
Flow Control
with BS, STmin
Consecutive Frame
STmin
Block
Size (BS)
Consecutive Frame
Consecutive Frame
Flow Control
with BS, STmin
Consecutive Frame
Consecutive Frame
last Block
(BS)

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3.2 Message Types

Each of the CAN message types shown above. i.e. Single Frame (SF), First Frame (FF), Consecutive Frame (CF) and Flow Control (FC) contain two main components, namely:

- Protocol Control Information (PCI):
  Indicates the frame type, sequence number counter for multiple frames, data length, Separation time (STmin), Block Size (BS) and Flow Status (FS).

- Transmission data (Data): Contains the data to be transmitted.

PCI and Transmission Data together make up the Protocol Data unit (PDU).

Data Field Structure					
FRAME TYPE	**BYTE 1**		**BYTE 2**	**BYTE 3**	**BYTES 4-8**
	Bits 7-4	Bits 3-0			
SINGLE FRAME (SF)	**0000**	**SF_DL**	DB 1	DB 2	DB 3 – 7
FIRST FRAME (FF)	**0001**	**FF_DL**		DB 1	DB 2 – 6
CONSECUTIVE FRAME (CF)	**0010**	**SN**	DB 1	DB 2	DB 3 – 7
FLOW CONTROL (FC)	**0011**	**FS**	BS	STmin	00

SF_DL : 4 Bit Single Frame Data length
FF_DL : 12 Bit First Frame Data length
SN : Sequence Number
FS : Status of data stream
BS : Block Size
STmin : Minimum Separation Time
DB : Data byte

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3.2.1 Single Frame

Single Frame message type is used for data blocks with a maximum data length of 7-Bytes and is identified by the upper nibble of PCI set to 0000b.

Definition of Single Frame data length (SF_DL):

Hex Value	Description
0 - 7	Single Frame Data length (SF_DL) is a 4-bit value that contains the number of data bytes present in the single frame

3.2.2 First Frame

First Frame message type is used for data blocks which have more than 7-Bytes. They are used in combination with one or more Consecutive Frames and is identified by the upper nibble of PCI set to 0001B.

Definition of First Frame Data length (FF_DL):

Hex Value	Description
000 – 007	Reserved
008 – FFF	First Frame Data length (FF_DL) contains number of data bytes.

3.2.3 Consecutive Frame

Consecutive Frames follows the First Frame and is identified by the upper nibble of PCI set to 0010b.

The lower nibble of the PCI has a Sequence Number (SN) which is determined as follows:

First Frame is regarded as containing Sequence Number 0. Hence the first Consecutive Frame always holds Sequence Number 1. Sequence Number is incremented by one for every additional Consecutive Frame. A Flow Control frame sent 'in between' does not affect the Sequence Number counting. When the Sequence Number reaches the value 15, the following Consecutive Frame shall be sent with Sequence Number set to 0.

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Definition of Sequence Number (SN):

Protocol Data name	FF	CF	CF	CF	CF	CF	CF	CF	CF
SN (hex)	0	1	2	...	E	F	0	1	...

Hex Value	Description
0 – F	Sequence Number (SN) The Sequence Number has a range of 0 to 15. SN-Value is stored in the lower nibble of Byte 1.

3.3 Flow Control

After the First Frame or the last Consecutive Frame of a block of frames (when further CFs are required to complete the data transfer), a Flow Control message shall be sent by the receiver node and is identified by the upper nibble of PCI set to 0011b. With the Flow Control, the receiving node notifies the transmitting node how many messages it shall receive between two Flow Control frames (maximum Block Size, Bsmax) and at what rate Consecutive Frames (minimum Separation Time, STmin). The lower nibble of the Flow Control PCI shall have a Flow Control Status (FS).

Definition of Flow Control Status (FS):

Hex Value	Description
0	Continue To Send (CTS) The Receiver is ready and is waiting for maximally BS Consecutive Frames
1	Wait (WT) The receiver is not ready to receive new messages and informs transmitter to wait for new FlowControl and to restart its N_BS timer. If FlowStatus is set to Wait, the values of BS and STmin in the FlowControl message are not relevant and shall be ignored.
2	Overflow (OVFLW) The FF_DL received in FF is greater than the buffer size then FS is set to overflow. If FlowStatus is set to Overflow, the values of BS and STmin in the FlowControl message are not relevant and shall be ignored.
3 - F	Reserved

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Block Size (Bsmax):

Hex Value	Description
00	Block Size (Bsmax) Indicates that no more Flow Control frames are allowed during transmission of the segmented data frame. All Consecutive Frames shall be sent without any stop for further Flow Control frames.
01 – FF	Block Size (Bsmax) Defines the maximum number of Consecutive Frames that shall be received without an intermediate Flow Control frame.

Separation Time (STmin):

Hex Value	Description
00 – FF	Separation Time (STmin) The Separation Time defines the minimal allowed time gap (in ms) between the transmissions of Consecutive Frames.

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4 Communication Parameters

Only one tester address is permitted in one diagnosis session all the time. A change of the tester address causes an ECU Timeout. The number of switches between functional and physical addressing is unlimited in one diagnosis session.

Name	Category	Value	Description
Bus type	Communication	CAN	Type of transport media
Baud rate	Communication	500000 bits/s	Bus Speed of the used transport media.
Addressing Scheme	Communication	Normal	Addressing Scheme used for physical/Functional requests and responses.
CAN-ID Type	Communication	11-bit	CAN-ID Type used for physical/Functional requests and responses
Max Length of TP Message	Communication	4095-Bytes	
Physical Request CAN-ID	Communication	0x7E1	The request CAN identifier for physical requests. The hex value not only describes the identifier but also the priority of the message. The higher the number, the lower the priority.
Response CAN-ID	Communication	0x7E9	The response CAN identifier for physical responses. The response for functional requests shall be sent via the physical path.
Functional Request CAN-ID	Communication	0x7DF	The request CAN identifier for functional requests.
P2Server	Timing	50ms	Performance requirement for the ECU to start with the response message after the reception of a request message.
P2*Server	Timing	5000ms	Performance requirement for the ECU to start with the response message after the transmission of a negative response message with response code 78 hex (enhanced response timing).
S3Server	Timing	5000ms	Time for the ECU to keep a diagnostic session other than the Default Session active while not receiving any diagnostic request message
STmin	Timing	127ms	(Separation Time) defines the minimum time gap between consecutive frames.
Blocksize	Communication	0	The number 0 tells the transmitter, that no more flow controls shall disrupt the sending of the remaining flow controls
Timeout As	Timing	50ms	Time for transmission of frame, transmitter side
Timeout Ar	Timing	50ms	Time for transmission of frame, receiver side
Timeout Bs	Timing	1000ms	Time until reception of next flow control
Time Br	Timing	100ms	Time for next transmission of flow control
Time Cs	Timing	800ms	Time until next transmission of consecutive frame
Timeout Cr	Timing	1000ms	Time until reception of next consecutive frame

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Name	Category	Value	Description
RC78CompletionTimeout	Timing	50000ms	Time period the tester accepts repeated negative responses with response code 0x78 and waits for a positive response further on. After sending 10 consecutive NRC78, Dcm sends NRC10 (generalReject).

5 Service Overview

The UDS services / functions supported by the ECU, together with the request code from tester and the corresponding response code from the ECU are listed below:

Service Identifier	Service Name	Addressing Mode		Positive Response
		Physical	Functional	
$10	DiagnosticSessionControl	x	x	$50
$11	ECUReset	x	-	$51
$14	ClearDiagnosticInformation	x	x	$54
$19	ReadDTCInformation	x	-	$59
$22	ReadDataByIdentifier	x	-	$62
$27	SecurityAccess	x	-	$67
$2E	WriteDataByIdentifier	x	-	$6E
$31	RoutineControl	x	-	$71
$28	CommunicationControl	x	x	$68
$85	ControlDTCSetting	x	x	$C5
$3E	TesterPresent	x	x	$7E

Note:

- x: Service is available in the specified addressing mode.
- -: Service is unavailable in the specified addressing mode.

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6 Availability of Services: Based on Session

The ECU shall execute a UDS service requested by the Tester depending on:

- The type of Diagnostic mode active at the time of request
- Vehicle speed at the time of request
- ECU being busy executing some other internal routine (e.g.: executing control sequence)

6.1 List of service availability based on the diagnostic session and security.

Service Name	Default	Extended	Programming
DiagnosticSessionControl			
Default Session	Y	Y	N
Programming Session	Y	N	Y
Extended Session	Y	Y	Y
EcuReset			
Hard Reset	N	Y	Y
Clear All DTC	Y	Y	N
ReadDtcInformation			
ReportNumberOfDTCByStatusMask	Y	Y	N
ReportDTCByStatusMask	Y	Y	N
SecurityAccess			
RequestSeed	Y	Y	Y
SendKey	Y	Y	Y
ReadDataByIdentifier			
ECU input signal	Y	Y	N
Variant Data	Y	Y	N
Reprogramming Date	Y	Y	N
Process Byte	Y	Y	N
Customer Part Number	Y	Y	N
Software Version	Y	Y	N
Software Fingerprint	Y	Y	Y
Vehicle Identification Number(VIN)	Y	Y	N
BBNumber	Y	Y	N
Diagnostic Active Session	Y	Y	Y
ReprogrammingInfo	Y	Y	Y
CC Feature Control	Y	Y	N
MTC Feature Control	Y	Y	N
Drag Torque Control	Y	Y	N
IMU Feature Control	Y	Y	N
EOL-Lean Angle Data	Y	Y	N
EOL-Pitch Angle Data	Y	Y	N
EOL Mode	Y	Y	N
IMU Sensor Status	Y	Y	N
EOL Expected Lean Angle	Y	Y	N
EOL Expected Pitch Angle	Y	Y	N
MTC Ride Mode	Y	Y	N
ABS Ride Mode	Y	Y	N
WriteDataByIdentifier			
ProcessByte	Y+	Y+	N
Vehicle Identification Number(VIN)	Y+	Y+	N
Variant Data	Y+	Y+	N
CC Feature Control	Y+	Y+	N

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MTC Feature Control	Y+	Y+	N
Drag Torque Control	Y+	Y+	N
IMU Feature Control	Y+	Y+	N
EOL Mode	Y+	Y+	N
EOL Expected Lean Angle	Y+	Y+	N
EOL Expected Pitch Angle	Y+	Y+	N
Routine Control			
Evac and Fill	N	Y+	N
Repair Bleed	N	Y+	N
Actuator test	N	Y+	N
Wheel Speed Sensor Test	N	Y+	N
ABS SILA Lamp Blinking Test	N	Y+	N
Sensor Mounting Validation Test	N	Y+	N
Communication Control	N	Y	N
Control DTC Setting	N	Y	N
Tester Present	Y1	Y	Y

Note:

- Y: The service shall be executed in this session.
- N: The service shall not be executed in this session.
- Y+: Protected services/function.
- Y1: The service is supported in this session without function i.e., session timer inactive.

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6.2 List of the supported sub-function

Service		Description
Service Identifier (hex)	**Sub-function (hex)**	
10		**DiagnosticSessionControl**
	01/81	DefaultSession
	02/82	ProgrammingSession
	03/83	ExtendedSession
11		**ECUReset**
	01/81	Hard Reset
19		**ReadDTCInformation**
	01	reportNumberOfDTCByStatusMask
	02	reportDTCByStatusMask
27		**SecurityAccess**
	01	RequestSeed
	02	SendKey
28		**CommunicationControl**
	00/80	EnableRxAndTx
	01/81	EnableRxAndDisableTx
31		**RoutineControl**
	01/81	StartRoutine
	02/82	StopRoutine
	03	RequestRoutineResults
85		**ControlDTCSetting**
	01/81	DTC storage enabled
	02/82	DTC storage disabled
3E		**TesterPresent**
	00/80	suppressPosRspMsgIndicationBit

SuppressPositiveResponse bit:

The most significant bit (MSB) in the sub-function byte of certain tester requests, is used to activate and deactivate the positive ECU response. This bit is called the SuppressPositiveResponse bit (SPR) and does not belong to the sub-function. Therefore, it is not contained in the positive response from the ECU.
Below the definition of the SuppressPosResponse:

SPR	Description
0b	The ECU shall send a positive or negative response.
1b	The ECU shall suppress the positive response. A positive response may be sent only if a negative response with NRC 0x78 has been sent beforehand.

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7 DiagnosticSessionControl

The DiagnosticSessionControl is used to enable different diagnostic sessions in the ECU. The current project supports the following sessions:

Default Session

This mode does not permit the execution of all diagnostic services possible for the ECU (see "Availability of Services" section for services supported during this session).
The ECU shall always start in default session with only one session being active at any given time.
The ECU shall only transition to a non-default session following a request from the diagnostic tester to do so.

Extended Diagnostic session

During this mode the ECU shall be switched to Extended Diagnostic session and all the controller functions shall be set to passive with only the diagnostic services set to active. All lamps shall be turned ON to indicate the non-availability of the controller.

During the Extended Diagnostic session, the ECU shall monitor the time taken for every new tester request. The ECU shall return to Normal / Default Diagnosis mode if this time exceeds S3Server (see "Communication Parameters" section). The extended session shall be entered only if vehicle speed is less than 10km/h.

Programming Session

If the reprogramming of the ECU is needed, it shall be set in this mode.

DiagnosticSessionControl Request:

Tester ⇒ ECU: StartDiagnosticSessionRequest		
Byte	Contents	Value
1	DiagnosticSessionControl	10 H
2	Sub-function	xx H

Definition of Sub-function:

No	Sub-function	Session Value
1	Default Session	01 H / 81 H
2	Programming Session	02 H / 82 H
3	ExtendedDiagnosticSession	03 H / 83 H

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Default Session Positive Response:

Positive response table for default session.

ECU ⇒ Tester: StartDiagnosticSessionPositiveResponse		
Byte	Contents	Value
1	StartDiagnosticSessionPR	50 H
2	DefaultSession	01 H
3	P2Server_max Timing HB	xx H
4	P2Server_max Timing LB	xx H
5	P2*Server_max Timing HB	xx H
6	P2*Server_max Timing LB	xx H

Programming Session Positive Response:

Positive response table for Programming Session

ECU ⇒ Tester: StartDiagnosticSessionPositiveResponse		
Byte	Contents	Value
1	StartDiagnosticSessionPR	50 H
2	Programming Session	02 H
3	P2Server_max Timing HB	xx H
4	P2Server_max Timing LB	xx H
5	P2*Server_max Timing HB	xx H
6	P2*Server_max Timing LB	xx H

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Extended Diagnostic Session Positive Response:

The ExtendedDiagnosticSession permit the execution of all diagnostic services possible for the ECU.

Positive response table for extended session.

ECU ⇒ Tester: StartDiagnosticSessionPositiveResponse		
Byte	Contents	Value
1	StartDiagnosticSessionPR	50 H
2	ExtendedDiagnosticSession	03 H
3	P2Server_max Timing HB	xx H
4	P2Server_max Timing LB	xx H
5	P2*Server_max Timing HB	xx H
6	P2*Server_max Timing LB	xx H

Note:

- P2Server_max is in 1 ms resolution.
- P2*Server_max is in 10ms resolution.
- If the suppressPosRspMsgIndicationBit is set, then the positive response from the ECU is suppressed.
- If a current diagnostic session has been requested that is already active, the ECU shall respond with a positive response and shall not have any effect on the current session or its available services and/or functionality.

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Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: DiagnosticSessionControlNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	DiagnosticSessionControl	10 H
3	NegativeResponseCode	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x12	Sub-FunctionNotSupported	This NRC shall be sent if the sub-function parameter is not supported.
0x22	ConditionsNotCorrect	This NRC shall be returned if the criteria for the request DiagnosticSessionControl are not met.
0x78	RequestCorrectlyReceived-ResponsePending	This NRC indicates that the request message was received correctly, but the action to be performed is not yet completed and the ECU is not yet ready to receive another request. As soon as the requested service has been completed, the ECU shall send a positive response message or negative response message with a response code different from this.
0x88	VehicleSpeedTooHigh	This NRC indicates that the requested action shall not be taken because the ECU prerequisite condition for vehicle speed is not met.

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8 ECU Reset

This service requests the ECU to effectively perform a ECU reset i.e.re-initialization of volatile and non-volatile memory. The ECUReset positive response message shall be sent before the reset is executed in the ECU.

ECU Reset Request:

Tester ⇒ ECU: ECUResetRequest		
Byte	Contents	Value
1	ECUReset	11 H
2	SubFunction	xx H

After a successful ECU reset the ECU shall activate the default session. During this time the ECU does not accept any request messages nor sends any response messages.

Definition of Sub function:

No	ECU Reset type	Session Value
1	Hard Reset	01 H / 81 H

Hard Reset Positive Response:

ECU shall send the following Positive response after successful Hard Reset.

Tester ⇒ ECU: ECU Reset PositveResponse		
Byte	Contents	Value
1	ECUResetPR	51 H
2	Hard Reset	01 H

Note:

- The ECUReset can be done only at vehicle speed less than 10 km/h. Otherwise, the ECU responds with a negative response code.
- In boot block there shall not be any speed check for this service.

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Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: ECU Reset Negative Response		
Byte	Contents	Value
1	Negative Response	7F H
2	ECU Reset	11 H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x12	Sub-functionNotSupported	This NRC shall be sent if the sub function parameter is not supported.
0x88	VehicleSpeedTooHigh	This NRC indicates that the requested action shall not be taken because the ECU prerequisite condition for vehicle speed is not met (current vehicle speed is above a pre-programmed maximum threshold).

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9 Clear Diagnostic Information

The "ClearDiagnosticInformation" service shall clear the contents of all failure memory stored in the EEPROM of the ECU. The ECU shall send a positive response even if no DTCs are stored.

Clear Diagnostic Information Request:

Tester ⇒ ECU : ClearDiagnosticInformationRequest		
Byte	Contents	Value
1	ClearDiagnosticInformation	14 H
2	GroupOfDTC HB	xx H
3	GroupOfDTC MB	xx H
4	GroupOfDTC LB	xx H

Definition of GroupOfDTC:

No	DID Description	DID value
1	All DTCs	GroupOfDTC HB: FF H GroupOfDTC MB: FF H GroupOfDTC LB: FF H

Positive Response:

On gaining access to the EEPROM resource, ECU shall send the following Positive response:

ECU ⇒ Tester: ClearDiagnosticInformationPositiveResponse		
Byte	Contents	Value
1	ClearDiagnosticInformationPR	54 H

Note:

- If a failure continues to persist in the system, it shall be logged after the ClearDiagnosticInformation request is given.

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Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: ClearDiagnosticInformationNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	ClearDiagnosticInformationRequest	14 H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x31	RequestOutOfRange	This NRC shall be returned if the specified groupOfDTC parameter is not supported.
0x78	ResponsePending	This NRC indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the ECU is not yet ready to receive another request.

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10 Read DTC Information

The “ReadDiagnosticCodesByStatus” service shall report all Diagnostic Trouble codes stored in the EEPROM. Please refer Failsafe document for the list of available DTCs.

Read DTC Information Request:

Tester ⇒ ECU : ReadDiagnosticTroubleCodesRequest		
Byte	Contents	Value
1	ReadDTC Information	19 H
2	SubFunction	xx H
3	DTCStatusAvailability Mask	xx H

Definition of Sub function:

No	Sub function Description	Sub function value
1	ReportNumberOfDTCByStatusMask	01 H
2	ReportDTCbyStatusMask	02 H

Definition of DTCStatusAvailability Mask: 0x2FH:

StatusOfDTC	Bit #	Bit State	Description
Test Failed	0	1	This bit is set to indicate a matured failing result.
TestFailedThisOperationCycle	1	1	This bit is set if the DTC was reported at least once during the current operating cycle.
PendingDTC	2	1	This bit is set and latched if a malfunction is detected during the current operating cycle.
ConfirmedDTC	3	1	This bit is set if the DTC is confirmed at least once since the last call to ClearDiagnosticInformation and if the aging criteria has not yet been satisfied.
TestNotCompletedSinceLastClear	4	0	This bit is set to indicate that the DTC test has not run to completion since the last time diagnostic information was cleared.
TestFailedSinceLastClear	5	1	This bit is set to indicate that the DTC test returned a failed result at least once since the last time diagnostic information was cleared.
TestNotCompletedThisOperationCycle	6	0	This bit is set to indicate that that the DTC test has not run to completion this operation cycle (or since the last time diagnostic information was cleared this operation cycle).
WarningIndicatorRequested	7	0	This bit is set to indicate that the ECU is requesting warningIndicator to be active

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10.1 ReportNumberOfDTCByStatusMask

This sub function specifies that the ECU shall transmit to the tester the number of DTCs matching the tester's specified status availability mask.

ReportNumberOfDTCByStatusMask Request:

Tester ⇒ ECU : ReadDiagnosticTroubleCodesRequest		
Byte	Contents	Value
1	ReadDiagnosticCodesByStatus	19 H
2	SubFunction	01 H
3	DTCStatusMask	xx H

Positive response:

ECU ⇒ Tester: ReadDiagnosticTroubleCodesPositiveResponse		
Byte	Contents	Value
1	ReadDTCInformationPR	59 H
2	SubFunction	01 H
3	DTCStatusAvailabilityMask	xx H
4	DTCFormatIdentifier (ISO_14229-1_DTCFormat)	01 H
5	DTCCountHighByte	xx H
6	DTCCountLowByte	xx H

Definition of DTC Count:

DTCCount parameter is a 2-byte unsigned numeric number containing the number of DTCs available in the ECU's memory based on the status mask provided by the tester.

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10.2 ReportDTCbyStatusMask

This parameter specifies that the ECU shall transmit to the tester a list of DTCs and corresponding Status matching a tester defined status mask.

ReportDTCbyStatusMask Request:

Tester ⇒ ECU : ReadDiagnosticTroubleCodesRequest		
Byte	Contents	Value
1	ReadDiagnosticCodesByStatus	19 H
2	SubFunction	02 H
3	DTCStatusMask	xx H

Positive response:

ECU ⇒ Tester: ReadDiagnosticTroubleCodesPositiveResponse		
Byte	Contents	Value
1	ReadDTCInformation PR	59 H
2	SubFunction	02 H
3	DTCStatusAvailabilityMask	xx H
4+(n*4)	DTC n HB	xx H
5+(n*4)	DTC n MB	xx H
6+(n*4)	DTC n LB	xx H
7+(n*4)	StatusOfDTC n	xx H

Note:

- n: Number of fault records (Current project supports 10 records).
- If no fault is present in the system, the following response shall be sent:

ECU ⇒ Tester: ReadDTCInformationPositiveResponse		
Byte	Contents	Value
1	ReadDTCInformation PR	59 H
2	SubFunction	02 H
3	DTCStatusAvailabilityMask	xx H

Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: ReadDiagnosticTroubleCodesNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	ReadDTC Information	19 H
3	Response Code	xx H

Response Code	Description	Reason
0x12	Sub-functionNotSupported	This NRC shall be sent if the sub-function parameter is not supported.
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x31	RequestOutOfRange	This NRC shall be sent if the tester specified a DTCMaskRecord that was not recognized by the ECU or an invalid DTCSnapshotRecordNumber.

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11 Security access

This service shall provide a means to access data and/or diagnostic services, which have restricted access for security, emissions, or safety reasons. The security access routine is required to authenticate the tester, enabling access to protected services. It also allows the ECU to enter Reprogramming Mode for re-flash process.

Security Access Request:

TESTER ⇒ ECU: SecurityAccessRequest		
Byte	Contents	Value
1	SecurityAccess Request SID	27 H
2	SecurityAccessType	xx H
3	SecurityAccessDataRecord [1]	xx H
	...	xx H
N	SecurityAccessDataRecord[m]	xx H

Definition of 'SecurityAccessType ':

No	Data Identifier	Value
1	Request Seed	01 H
2	Send Key	02 H

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11.1 Request Seed

On reception of this command, ECU generates a random number known as Seed (2-Byte value) and sends it as response to the tester.

Request seed command is shown below:

TESTER ⇒ ECU: Request Seed Command		
Byte	Contents	Value
1	SecurityAccess Request SID	27 H
2	SecurityAccessType	01 H

The seed generated shall always be a non-zero value. However, in case the security has already been unlocked, the seed returned shall be 0000H.This provides an indication to the tester that the current security level is in the unlocked state.

On receiving valid request, the ECU generates a 2-Byte seed and sends it as a positive response as shown below:

ECU ⇒ TESTER: Send Seed Request Positive Response		
Byte	Contents	Value
1	SecurityAccess Request PR	67 H
2	SecurityAccessType	01 H
3	Seed (HB)	xx H
4	Seed (LB)	xx H

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11.2 Send Key

In SendKey request, the tester sends a 2-Byte authentication key back to the ECU. The SecurityAccessType byte shall always be one more than that requested in the RequestSeed service.

TESTER ⇒ ECU: Send Key Request		
Byte	Contents	Value
1	SecurityAccess Request SID	27 H
2	SecurityAccessType	02 H
3	Key (HB)	xx H
4	Key (LB)	xx H

If the key sent to the ECU matches with that computed internally, a positive response shall be sent to indicate that the requested level of security has been unlocked.

ECU ⇒ TESTER: Send Key Positive Response		
Byte	Content	Value
1	SecurityAccess Request PR	67 H
2	SecurityAccessType	02 H

Note:

- Every time an unsuccessful attempt to unlock the ECU is detected, an error counter shall be incremented and shall reach a maximum value of 05 in one ignition cycle. The counter shall be reset every time the security is cancelled successfully.

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Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ TESTER: Security Access Negative Response		
Byte	Contents	Value
1	Negative Response	7F H
2	Requested Service: “Security Access Request SID “	27 H
3	Response Code	xx H

Response Code	Description	Reason
0x12	Sub-functionNotSupported	This NRC shall be sent if the sub-function parameter is not supported.
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x22	Conditions not correct	This NRC shall be returned if the criteria for the request SecurityAccess are not met.
0x24	Request sequence error	Sent if the ‘SendKey’ sub-function is received without first receiving a ‘RequestSeed’ request message.
0x35	Invalid Key	Sent if an expected 'SendKey' sub-function value is received and the value of the key does not match the ECU’s internally stored/calculated key.
0x36	Exceed number of attempts	Sent if the delay timer is active due to exceeding the maximum number of allowed false access attempts.
0x37	Required time delay not expired	Sent if the delay timer is active and a request is transmitted.

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12 Read Data by Identifier

The ReadDataByIdentifier service allows the tester to request data record values from the ECU identified by one or more dataIdentifiers. This service shall report the system information data. The format and definition of the data record shall be vehicle manufacturer/system supplier specific, and may include analog input and output signals, digital input and output signals, internal data, and system status information, if supported by the ECU.

Request for ReadDataByIdentifer:

TESTER ⇒ ECU: ReadDataByLocalIdentifierRequest		
Byte	Contents	Value
1	ReadDataByIdentifier	22 H
2	Data Identifier HB	xx H
3	Data Identifier LB	xx H

Definition of Supported Data Identifier:

No.	Data Identifier description	Data Identifier Value
1	ECU Input signal	0xF102 H
2	Variant Data *	0xF1FF H
3	Reprogramming Date *	0xF199 H
4	ProcessByte *	0xF1FA H
5	Customer Part Number **	0xF187 H
6	Software Version	0xF181 H
7	Software Fingerprint **	0xF184 H
8	Vehicle Identification Number (VIN) **	0xF190 H
9	BBNumber	0xF194 H
10	ReprogrammingInfo	0XF185 H
11	Diagnostic Active Session	0xF186 H
12	CC Feature Control *	0x3000 H
13	MTC Feature Control *	0x3001 H
14	Drag Torque Control*	0x3002 H
15	IMU Feature Control*	0x3006 H
16	EOL-Lean Angle Data	0xF108 H
17	EOL-Pitch Angle Data	0xF10A H
18	EOL Mode	0xF1F1 H
19	IMU Sensor Status	0xF107 H

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20	EOL Expected Lean Angle	0xF1F2 H
21	EOL Expected Pitch Angle	0xF1F5 H
22	MTC Ride Mode	0x2FFE H
23	ABS Ride Mode	0x2FFF H

Note:

- In the current implementation, if the data element of the requested data identifier is empty i.e., there is no data explicitly filled into the EEPROM, positive response is obtained with output buffer filled with 0x30H.
- * Indicates that for the specified service, if the EEPROM is empty, NRC22 is triggered.
- ** Indicates that for the specified service, if the EEPROM is empty, the output buffer is filled with 0x00 as the mentioned services output ASCII values.

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12.1 ECU Input signal

This service shall be used to read the wheel speeds, the power supply voltage and the status of the valves during actuations.
This service shall be read out while routine control services like Evac and Fill Repair Bleed and actuator test are being executed to check the valves being actuated.

On a valid request, the ECU shall send the following Positive response:

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifier HB	F1 H
3	Data Identifier LB	02 H
4	Front Wheel Speed (1 LSB = 1 Km/h)	xx H
5	Rear Wheel Speed (1 LSB = 1 Km/h)	xx H
6	Power Supply Voltage (1 LSB = 0.078 V)	xx H
7	Valve status 1 * -Bit0: Valve relay status, 1=ON -Bit1: Pump motor status, 1=ON -Bit2: Front Inlet Valve status, 1=ON -Bit3: Front Outlet Valve status, 1=ON -Bit4: Reserved -Bit5: Reserved -Bit6: Rear Inlet Valve status, 1=ON -Bit7: Rear Outlet Valve status, 1=ON	xx H
8	Valve status 2 * -Bit0 to Bit7: Reserved	xx H

Note:

- Byte8 shall return 0x00H.
- If speed greater than 0xFFH is provided as input, Bytes 4 and 5 shall truncate it to a max value of FFH.

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Definition of Valve Status:

*: Relationship-Table between Inlet/Outlet Valve State and Braking Pressure

	Inlet Valve (EV)	Outlet Valve (AV)
Pressure Hold	1	0
Pressure Decrease	1	1
Pressure Increase	0	0

The valve activation is limited by overheat protection measures.

12.2 Variant Data

The tester uses this request to read the variant data written in the EEPROM.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifier Response		
Byte	Content	Value
1	ReadDataByIdentifierPR	62 H
2	DataIdentifier (HB)	F1 H
3	DataIdentifier (LB)	FF H
4	Variant byte	xx H

Note:

- TVS U400 - Variant supported value is 0x01.
- TVS U490 - Variant supported value is 0x02.
- TVS U712/ U713 - Variant supported value is 0x04 and 0x05.

12.3 Reprogramming Date

The service is used to read the date of reprogramming of ECU. On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Content	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifier HB	F1 H
3	Data Identifier LB	99 H
4	Programming Date Year (YY) HB	xx (BCD)
5	Programming Date Month (MM)	xx (BCD)
6	Programming Date Day (DD) LB	xx (BCD)

Note:

- Reprogramming Date is available only if Boot Block feature is available and is configured in boot block or programming mode.

12.4 ProcessByte

Process byte is used to check status of evacuation and fill routine.

On a valid request, the ECU shall send the following positive response:

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifier HB	F1 H
3	Data Identifier LB	FA H
4	EOL Assembly process byte	xx H

Note:

- EOL Assembly process byte: 01 (Completed: Wet state)
- EOL Assembly process byte: 00 (Initial state: Dry state)

12.5 Customer part number

8-Byte Customer part number shall be read out using this service.

On a valid request, the ECU shall send the following positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifier HB	F1 H
3	Data Identifier LB	87 H
4	Customer part number (ASCII)	xx H
5		xx H
6		xx H
7		xx H
8		xx H
9		xx H
10		xx H
11		xx H

Note:

- The ASCII values shall be read from the customer tester tool.

12.6 Software Version

This service shall be used to read out the software version. Platform supports a 2-Byte read to indicate the type of software release done.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifier HB	F1 H
3	Data Identifier LB	81 H
4	SoftwareVersion_Major	xx H
5	SoftwareVersion_Minor	xx H

Definition of Software Version:

SoftwareVersion_Major: Sample release to customer (C/D sample or bugfix/running change).
SoftwareVersion_Minor: Intermediate and internal releases.

12.7 Software Fingerprint

This service shall be used to read the 12-Byte Software Fingerprint.

On a valid request, the following positive response shall be sent:

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Content	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	84 H
4	Reprogram Part Number Position #1 (HB)	ASCII
5	Reprogram Part Number Position # 2	ASCII
6	Reprogram Part Number Position # 3	ASCII
7	Reprogram Part Number Position # 4	ASCII
8	Reprogram Part Number Position # 5	ASCII
9	Reprogram Part Number Position # 6	ASCII
10	Reprogram Part Number Position # 7	ASCII
11	Reprogram Part Number Position # 8	ASCII
12	Reprogram Part Number Position # 9	ASCII
13	Reprogram Part Number Position #10	ASCII
14	Reprogram Part Number Position #11	ASCII
15	Reprogram Part Number Position #12(LB)	ASCII

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12.8 Vehicle Identification Number

17-Byte VIN (Vehicle Identification Number) shall be read using this service.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	90 H
4	VIN Data (ASCII)	ASCII
5		ASCII
6		ASCII
7		ASCII
8		ASCII
9		ASCII
10		ASCII
11		ASCII
12		ASCII
13		ASCII
14		ASCII
15		ASCII
16		ASCII
17		ASCII
18		ASCII
19		ASCII
20		ASCII

Note:

- ASCII values are read from EEPROM.

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12.9 BBNumber

5-Byte of Bosch software number shall be read using this service. This data is obtained directly from the flash.

On a valid request, the ECU shall send the following Positive response:

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse			
Byte	Contents		Value
1	ReadECUIdentificationPR		62H
2	Data Identifier HB		F1 H
3	Data Identifier LB		94 H
4	SW BB-Number (ASCII)	Byte 1	xx H
5		Byte 2	xx H
6		Byte 3	xx H
7		Byte 4	xx H
8		Byte 5	xx H

12.10 Diagnostic Active Session

1-Byte is allocated to read the currently active Diagnostic session.

On a valid request, the ECU shall send the following Positive response:

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	86 H
4	DiagnosticActiveSession	xx H

Note:

- A DiagnosticActiveSession when control is different layers is mentioned in below table:

Application	Boot
0x01 (Default Session)	0x81 (Default Session)
0x03 (Extended Session)	0x82 (Programming Session)
-	0x83 (Extended Session)

12.11 ReprogrammingInfo

This service allows the read of all the reprogramming information written into the EEPROM in the programming session.

On a valid request, the ECU shall send the following Positive response:

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Content	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifier HB	F1 H
3	Data Identifier LB	85 H
4	Programming Date Year (YY)	xx (BCD)
5	Programming Date Month (MM)	xx (BCD)
6	Programming Date Day (DD)	xx (BCD)
7	Reprogram Part Number Position #1 (HB)	xx (ASCII)
8	Reprogram Part Number Position # 2	xx (ASCII)
9	Reprogram Part Number Position # 3	xx (ASCII)
10	Reprogram Part Number Position # 4	xx (ASCII)
11	Reprogram Part Number Position # 5	xx (ASCII)
12	Reprogram Part Number Position # 6	xx (ASCII)
13	Reprogram Part Number Position # 7	xx (ASCII)
14	Reprogram Part Number Position # 8	xx (ASCII)
15	Reprogram Part Number Position # 9	xx (ASCII)
16	Reprogram Part Number Position #10	xx (ASCII)
17	Reprogram Part Number Position #11	xx (ASCII)
18	Reprogram Part Number Position #12(LB)	xx (ASCII)
19	BBNumber(HB)	xx
20	BBNumber	xx
21	BBNumber(LB)	xx

Example:

Reprogramming Date: 19 04 12 (yy mm dd)
BB Number: 09 95 96 (BB99596)
SoftwareFingerPrint: 12 Byte ASCII value (123456789ABC)

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12.12 CC Feature Control

The tester uses this request to read the CC (Cruise Control) Feature status.

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	30 H
3	Data Identifiers LB	00 H
4	CC Feature Control Byte	xx H

Note:

- CC Feature Control Byte - 0x00 (disable)
  0x01 (enable)

12.13 MTC Feature Control

The tester uses this request to read the MTC (Motorcycle Traction Control) Feature status.

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	30 H
3	Data Identifiers LB	01 H
4	MTC Feature Control Byte	xx H

Note:

- MTC Feature Control Byte - 0x00 (disable)
  0x01 (enable)

12.14 Drag Torque Control

The tester uses this request to read the DTC (Drag Torque Control) Feature status.

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	30 H
3	Data Identifiers LB	02 H
4	DTC Feature Control Byte	xx H

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Note:

- DTC Feature Control Byte - 0x00 (disable)
  0x01 (enable)

12.15 IMU Feature Control

The tester uses this request to read the IMU Feature status.

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	30 H
3	Data Identifiers LB	06 H
4	IMU Feature Control Byte	xx H

Note:

- IMU Feature Control Byte - 0x00 (disable)
  0x01 (enable)

12.16 EOL-Lean Angle Data

This service is used to read the learnt lean angle values. This data element gets updated on running the routine "Sensor Mounting Validation Test".
The learnt value is computed by taking an average of 10 lean angle values from the sensor and compared with the expected value (stored in EEPROM).

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	08 H
4	Lean Angle Data (HB)	xx H
5	Lean Angle Data (LB)	xx H

Note:

- Angle is read in degrees in signed word format with a resolution of 1 Degree per bit.

SL No	Angle	HB	LB
1	Positive Values	XX	XX
2	Negative Values	FX	XX
3	0	00	00
4	180	00	B4
5	-180	FF	4C

12.17 EOL-Pitch Angle Data

This service is used to read the learnt pitch angle values. This data element gets updated on running the routine "Sensor Mounting Validation Test".
The learnt value is computed by taking an average of 10 pitch angle values from the sensor and compared with the expected value (stored in EEPROM).

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	0A H
4	Pitch Angle Data (HB)	xx H
5	Pitch Angle Data (LB)	xx H

Note:

- Angle is read in degrees in signed word format with a resolution of 1 Degree per bit.

SL No	Angle	HB	LB
1	Positive Values	XX	XX
2	Negative Values	FX	XX
3	0	00	00
4	180	00	B4
5	-180	FF	4C

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12.18 EOL Mode

The phase information of the EOL shall be read through this request.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	F1 H
4	EOL Mode	xx H

For the successful running of "Sensor Mounting Validation Test" routine, this data element shall be initialized with value 0x01H.

Once the routine finishes, based on the status of completion the data element EOL Mode is updated.

The table below specifies the significance of the values read using this service:

EOL Mode Byte	EOL Mode Description
0x00	Successful Completion of EOL
0x01	Phase-1 of EOL Entered
0x02	EOL has entered the lock mode

Note:

- 0x02H indicates that the computed value of pitch/lean angle is not within the range (2 degrees for pitch angle and 4 degrees for lean angle) of the expected values.

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12.19 IMU Sensor Status

This service is used to read the lean and pitch angle values along with the IMU sensor status availability.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	07 H
4	Lean Angle LB	xx H
5	Lean Angle HB	xx H
6	Pitch Angle LB	xx H
7	Pitch Angle HB	xx H
8	IMU Sensor Status Availability Bit0: Sensor Availability, 1= Available, 0= Not available Bit1to Bit7: Reserved	xx H

Note:

- Angle is read in degrees in signed word format with a resolution of 1 Degree per bit.

SL No	Angle	HB	LB
1	Positive Values	xx	xx
2	Negative Values	Fx	xx
3	0	00	00
4	180	00	B4
5	-180	FF	4C

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12.20 EOL Expected Lean Angle

Lean Angle data that is expected during EOL for sensor calibration is read from EEPROM.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	F2 H
4	Lean Angle HB	xx H
5	Lean Angle LB	xx H

12.21 EOL Expected Pitch Angle

Pitch Angle data that is expected during EOL for sensor calibration is read from EEPROM.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	F1 H
3	Data Identifiers LB	F5 H
4	Pitch Angle HB	xx H
5	Pitch Angle LB	xx H

Note:

- The above two values are written into the EEPROM using the WriteDataByIdentifier service.

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12.22 MTC Ride Mode

This service is used to read the MTC ride mode set in the system.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	2F H
3	Data Identifiers LB	FE H
4	MTC Ride Mode Data	xx H

Note:

- Here, MTC Ride Mode data:
  0x00: Mode1
  0x01: Mode2

12.23 ABS Ride Mode

This service is used to read the ABS ride mode set in the system.

On a valid request, the ECU shall send the following Positive response,

ECU ⇒ TESTER: ReadDataByIdentifierPositiveResponse		
Byte	Contents	Value
1	ReadDataByIdentifierPR	62 H
2	Data Identifiers HB	2F H
3	Data Identifiers LB	FF H
4	ABS Ride Mode Data	xx H

Note:

- Here, ABS Ride Mode data:
  0x01: Mode1
  0x02: Mode2
  0x03: Mode3
  0x04: Mode4

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Negative Response

Negative response code shall be in the below format.

ECU ⇒ Tester: ReadDataByIdentifierNegativeResponse		
Byte	Contents	Value
1	Negative Response	7FH
2	ReadDataByIdentifier	22H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalid Format	This NRC shall be sent if the length of the request message sent is wrong.
0x31	RequestOutOfRange	Requested data Identifier values are not supported by the ECU.
0x10	GeneralReject	This NRC indicates that the requested action has been rejected by the ECU.
0x78	RequestCorrectlyReceived-ResponsePending	This NRC indicates that the request message was received correctly, but the action to be performed is not yet completed and the ECU is not yet ready to receive another request. As soon as the requested service has been completed, the ECU shall send a positive response message or negative response message with a response code different from this.
0x7F	ServiceNotSupportedInActiveSession	This NRC indicates that the requested action shall not be taken because the ECU does not support the requested service in the session currently active. NRC 7F is obtained if read is done in functional addressing mode.

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13 Write Data by Identifier

The 'WriteDataByIdentifier' service shall store specified record data to EEPROM locations.

WriteDataByIdentifier Request:

Tester ⇒ ECU: WriteDataByIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifierRequest	2EH
2	Data Identifier HB	xx H
3	Data Identifier LB	xx H
4	Data Byte 1	xx H
--	--	xx H
n+3	Data Byte n	xx H

Definition of Data Identifier:

No	Data Identifier description	Data Identifier Value
1	ProcessByte	0xF1FA H
2	VehicleIdentificationNumber (VIN)	0xF190 H
3	Variant Data	0xF1FF H
4	CC Feature Control	0x3000 H
5	MTC Feature Control	0x3001 H
6	Drag Torque Control	0x3002 H
7	IMU Feature Control	0x3006 H
8	EOL Mode	0xF1F1 H
9	EOL Expected Lean Angle	0xF1F2 H
10	EOL Expected Pitch Angle	0xF1F5 H

Note:

- Ignition OFF/ON cycle is required after every EEPROM write cycle to get the data reflected in EEPROM.

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13.1 Process Byte

Process byte is used to check status of evac and fill routine.

The process byte value shall be written into the EEPROM using the below request:

Tester ⇒ ECU: WriteDataByIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	FA H
4	RecordValue	xx H

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifierPR	6E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	FA H

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13.2 Vehicle Identification Number

VIN data is written using below service.

Tester ⇒ ECU: WriteDataByIdentifierRequest		
Byte	Contents	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	90 H
4	VIN data (ASCII)	ASCII
5		ASCII
6		ASCII
7		ASCII
8		ASCII
9		ASCII
10		ASCII
11		ASCII
12		ASCII
13		ASCII
14		ASCII
15		ASCII
16		ASCII
17		ASCII
18		ASCII
19		ASCII
20		ASCII

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifierPR	6E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	90 H

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13.3 Variant Data

This service used to write variant in EEPROM.

Tester ⇒ ECU: WriteDataByLocalIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	F1 H
2	Data Identifier LB	FF H
3	Variant Data	xx H

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	FF H

Note:

- TVS U400 - Variant supported value is 0x01.
- TVS U490 - Variant supported value is 0x02.
- TVS U712/ U713 - Variant supported value is 0x04 and 0x05.

DRAFT

13.4 CC Feature Control

This service is used to enable or disable CC (Cruise Control) Feature.

Tester ⇒ ECU: WriteDataByLocalIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	30 H
2	Data Identifier LB	00 H
3	CC Feature Control Byte	xx H

Note:

- CC Feature Control Byte - 0x01 – Enable.
  0x00 – Disable.

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	30 H
3	Data Identifier LB	00 H

Note:

- NRC 31 shall be sent for xx values other than 0x00 and 0x01.
- NRC 22, Conditions not correct shall be sent if MTC is Off/ disabled or if CC is not supported in EMS variant data or if CC is not supported in Cluster variant data.

DRAFT

13.5 MTC Feature Control

This service is used to enable or disable MTC (Motorcycle Traction Control) Feature.

Tester ⇒ ECU: WriteDataByLocalIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	30 H
2	Data Identifier LB	01 H
3	MTC Feature Control Byte	xx H

Note:

- MTC Feature Control Byte - 0x01 – Enable.
  0x00 – Disable.

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	30 H
3	Data Identifier LB	01 H

Note:

- NRC 31 shall be sent for xx values other than 0x00 and 0x01.
- NRC 22, Conditions not correct shall be sent if TC is not supported in EMS variant data or if TC is not supported in Cluster variant data.

DRAFT

13.6 Drag Torque Control

This service is used to enable or disable DTC (Drag Torque Control) Feature.

Tester ⇒ ECU: WriteDataByLocalIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	30 H
2	Data Identifier LB	02 H
3	DTC Feature Control Byte	xx H

Note:

- DTC Feature Control Byte - 0x01 – Enable.
  0x00 – Disable.

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	30 H
3	Data Identifier LB	02 H

Note:

- NRC 31 shall be sent for xx values other than 0x00 and 0x01.
- NRC 22, Conditions not correct shall be sent if DTC is not supported in EMS variant data or if DTC is not supported in Cluster variant data.

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13.7 IMU Feature Control

This service is used to enable or disable IMU Feature.

Tester ⇒ ECU: WriteDataByLocalIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	30 H
2	Data Identifier LB	06 H
3	IMU Feature Control Byte	xx H

Note:

- IMU Feature Control Byte - 0x01 – Enable.
  0x00 – Disable.

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	30 H
3	Data Identifier LB	06 H

Note:

- NRC 31 shall be sent for xx values other than 0x00 and 0x01.
- This DID is also used to enable LC Feature Control.

DRAFT

13.8 EOL Mode

Service used to write EOL Mode in EEPROM. To run the EOL Routine "Sensor Mounting Validation Test", it is essential for this data element to be initialised with the value 0x01H.
The corresponding EEPROM shall be written into using the below service:

Tester ⇒ ECU: WriteDataByIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	F1 H
2	Data Identifier LB	F1 H
3	EOL Mode Byte	xx H

Definition of EOL Mode Byte:

EOL Mode byte	EOL Mode Description
0x00	To Exit from Lock state of EOL
0x01	To Enter Phase-1 of EOL

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	F1 H

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13.9 EOL Expected Lean Angle

This request is used to write the expected lean angle value into the corresponding data element and is essential for the "Sensor Mounting Validation Test" routine.

Tester ⇒ ECU: WriteDataByIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	F2 H
4	Lean Angle (HB)	xx H
5	Lean Angle (LB)	xx H

Note:

- Angle is read in degrees in signed word format with a resolution of 1 Degree per bit.

Sl No	Angle	HB	LB
1	Positive Values	xx	xx
2	Negative Values	Fx	xx
3	0	00	00
4	180	00	B4
5	-180	FF	4C

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	F2 H

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13.10 EOL Expected Pitch Angle

This request is used to write the expected pitch angle value into the corresponding data element and is essential for the "Sensor Mounting Validation Test" routine.

Tester ⇒ ECU: WriteDataByLocalIdentifierRequest		
Byte	Content	Value
1	WriteDataByIdentifier	2E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	F5 H
4	Pitch Angle (HB)	xx H
5	Pitch Angle (LB)	xx H

Note:

- Angle is read in degrees in signed word format with a resolution of 1 Degree per bit.

Sl No	Angle	HB	LB
1	Positive Values	XX	XX
2	Negative Values	FX	XX
3	0	00	00
4	180	00	B4
5	-180	FF	4C

On successful completion of the write process, the following positive response shall be sent:

ECU ⇒ Tester: WriteDataByIdentifierPositiveResponse		
Byte	Content	Value
1	WriteDataByIdentifier PR	6E H
2	Data Identifier HB	F1 H
3	Data Identifier LB	F5 H

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Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ TESTER: WriteDataByIdentifierNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	WriteDataByIdentifier	2E H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x31	RequestOutOfRange	This NRC shall be sent if: 1.The data Identifier in the request message is not supported in the ECU or the data Identifier is supported for read only purpose (via ReadDataByIdentifier service). 2.Any data transmitted in the request message after the data Identifier is invalid (if applicable to the node);
0x33	SecurityAccessDenied	This NRC shall be sent if the dataIdentifier, which reference a specific address, is secured and the ECU is not in an unlocked state.
0x10	GeneralReject	This NRC indicates that the requested action has been rejected by the ECU.
0x78	RequestCorrectlyReceived-ResponsePending	This NRC indicates that the request message was received correctly, but the action to be performed is not yet completed and the ECU is not yet ready to receive another request. As soon as the requested service has been completed, the ECU shall send a positive response message or negative response message with a response code different from this.

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14 Routine Control

The RoutineControl service is used by the tester to execute a defined sequence of steps and obtain any relevant results. This service shall start a ROM resident routine in the ECU and is run in the Extended session.

RoutineControl Request:

Byte	Contents	Value
1	RoutineControl Request SID	31 H
2	RoutineControlType	xx H
3	RoutineIdentifierHighByte	xx H
4	RoutineIdentifierLowByte	xx H
...	RoutineControlOptionRecord	xx H

Definition of RoutineControlType:

No	Sub Function description	SubFunctionValue
1	Start Routine	01 H/ 81 H
2	Stop Routine	02 H/ 82 H
3	RequestRoutineResults	03 H

Definition of 'RoutineIdentifiers':

No	Routine Identifier description	RoutineIdentifier Value
1	Evacuation and Fill	B001 H
2	Repair Bleed	B002 H
3	Actuator test	B003 H
4	Wheel Speed Sensor Test	B004 H
5	ABS SILA Blinking Test	B005 H

Note:

- This routine is applicable only if IMU Sensor is connected in the system.

DRAFT

14.1 Evac and Fill

Purpose:

During the vehicle production, no air shall remain in the brake system of the bike. After mounting of all hydraulic parts, an evacuation pump sucks the air out of the brake system. If the vacuum reaches defined pressure, the filling machine switches off the vacuum pump and starts to fill the bike with brake fluid. A valve actuation profile ensures that no air remains in the hydraulic ABS unit.

Entry and Exit conditions:

Entry condition:

The routine shall not start if any of the below specified conditions are met:

1. Failures that shut down the hydraulics (refer Fail-Safe specification).
2. Over Voltage (16.5V) and Under voltage (9.6V).
3. Other routine is running.

Exit condition:

The ongoing routine comes to a stop if the below conditions are encountered:

1. Failures that shut down the hydraulics (refer Fail-Safe specification).
2. Over Voltage (16.5V) and Under voltage (9.6V).

Process:

Based on the system, the corresponding valves are actuated on giving the Evac and Fill request. In this system both the front and rear outlet valves are actuated. The pump motor is actuated for the system based on the actuation pattern.

Note:

- Speed check is done only as entry condition into extended session and shall not be the entry/exit condition for routine.

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Start Routine Request

The request of Evacuation & Fill routine is shown here:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	01 H
5	ValveControlStatus (0xFF- ON)	xx H
6	TotalEvacAndFillTime (1 LSB = 1s; Max Limit :150s)	xx H
7	PumpControlStatus (0x00 – OFF; 0xFF- ON)	xx H
8	MotorRunTime (Time till Start of Pump motor from Start of routine Resolution: 1s)	xx H

Note:

- ValveControlStatus shall be made 0xFFH for valve actuations. 0x00H is an invalid value since EvacAndFill requires valve actuations.
- Maximum TotalEvacAndFillTime should be 150 seconds and MotorRunTime must not exceed 30 seconds to protect the routine from overheat failure.

Positive Response:

On a valid request and no failures (mentioned above), the ECU shall send the following Positive response.

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	01 H

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Stop Routine Request:

To stop the Evac and Fill process, the below request shall be sent:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	01 H

Positive Response:

On a valid request, the ECU shall send the following Positive response and ongoing routine is stopped.

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	01 H

Results Routine Request:

The status of the Evac and Fill routine shall be checked using the below request:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	01 H

Positive Response:

On a valid request, a positive response indicating the state of the routine requested is sent:

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	01 H
5	Routine Status	xx H

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Definition of Routine Status:

Value	Status
0x01	Routine is running
0x02	Routine has successfully finished
0x04	Routine has been stopped
0x08	Routine has been interrupted due to an error.

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14.2 Repair Bleeding

Purpose:

All systems that failed the on-line evacuation and fill process or might be replaced by dry units, shall be subject to the Repair-Bleed Routine.

The operator shall open the bleeding screws on the wheels and the system shall be filled by brake fluid.

Entry and Exit conditions:

Entry condition:

The routine shall not start if any of the below specified conditions are met:

1. Failures that shut down the hydraulics (refer Fail-Safe specification).
2. Over Voltage (16.5V) and Under voltage (9.6V).
3. Other routine is running.

Exit condition:

The ongoing routine comes to a stop if the below conditions are encountered:

1. Failures that shut down the hydraulics (refer Fail-Safe specification).
2. Over Voltage (16.5V) and Under voltage (9.6V).

Process:

Based on the phase requested, Front/Rear Outlet and the pump motor gets actuated based on the Repair Bleed pattern specified. There are a total of four phases that are defined.

Note:

- Speed check is done only as entry condition into extended session and shall not be the entry/exit condition for routine.

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Start Routine Request:

The request of Repair bleed routine is shown here:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	02 H
5	RepairBleedPhaseIdentifier	xx H
6	Reserved	xx H

Note:

- RepairBleedPhaseIdentifier specifies the phase of the repair bleed sequence and shall take values from 1 to 4.
- Valve and pump motor actuates in phase 2 and phase 4 for repair bleed for this system.
- The 6th byte is 'Reserved' bit is set as dummy and shall not have any impact on any of the phases as no repetition allowed for any phase.

Positive Response:

On receiving a valid request and no failures (mentioned above), the ECU shall send the following Positive response:

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	02 H

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Stop Routine Request:

Request to stop Repair bleed routine is as given below:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	02 H

Positive Response:

On receiving valid stop routine request, below response shall be sent.

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	02 H

Routine Results Request:

The status of the Repair Bleed routine shall be checked using the request result as shown below:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	02 H

Positive Response:

On receiving a valid RequestRoutineResults request, below response shall be sent.

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	02 H
5	Routine Status	xx H

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Definition of Routine Status:

Value	Status
0x01	Routine is running
0x02	Routine has successfully finished
0x04	Routine has been stopped
0x08	Routine has been interrupted due to an error.

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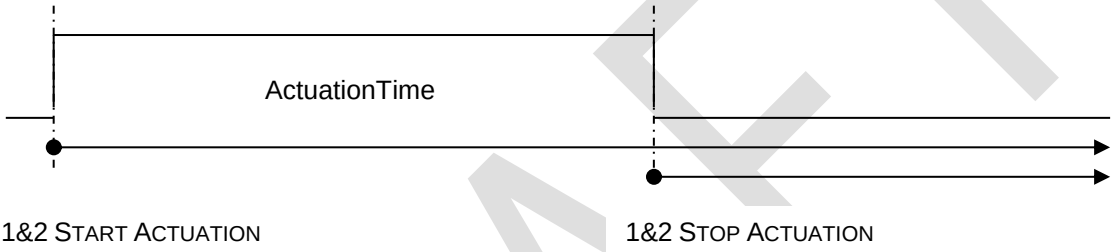
14.3 Actuator Test

The Actuator Test is a routine that gives the possibility to activate dedicated actuators (e.g., inlet valve / outlet valve). For each actuator the actuator number and the actuator value need to be set. This routine allows for actuation of valves and pump motor.

The normal use case of the Actuator test is to enable Actuators 1 and 2 for a certain time equal to the Actuation time. After this time the same actuators are switched off in the request by requesting OFF through 3/4.

The software has a self-protection mechanism (Diagnostic Hydraulic Protection- DHP) which shall set a failure when the time limit is exceeded, ensuring that the valves and motors are not damaged.
The actuator numbers and the values to enable / disable them are available in the diagnostic request.

Refer Appendix A for the list of actuators available.



Entry and Exit conditions:

Entry condition:

The routine shall not start if any of the below specified conditions are met:

1. Failures that shut down the hydraulics (refer Fail-Safe specification).
2. Over Voltage (16.5V) and Under voltage (9.6V).
3. Other routine is running.

Exit condition:

The ongoing routine comes to a stop if the below conditions are encountered:

1. Failures that shut down the hydraulics (refer Fail-Safe specification).
2. Over Voltage (16.5V) and Under voltage (9.6V).

Note:

- Speed check is done only as entry condition into extended session and shall not be the entry/exit condition for routine.

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Start Routine Request:

The Request of Actuator routine is shown here:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	03 H
5-6	1.Actuation Value	xxxx H
7-8	1.Actuator Number	xxxx H
9-10	2.Actuation Value	xxxx H
11-12	2.Actuator Number	xxxx H
13-14	Actuation Time (1LSB = 10ms, Max limit: 30s)	xxxx H
15-16	3.Actuation Value	xxxx H
17-18	3.Actuator Number	xxxx H
19-20	4.Actuation Value	xxxx H
21-22	4.Actuator Number	xxxx H

Positive Response:

On a valid request and no failures (mentioned above), the ECU shall send the following Positive response:

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	03 H

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Stop Routine Request:

The Actuator Test shall be stopped in between by a stop routine request.

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	03 H

Positive Response:

On receiving valid stop routine request, below response shall be sent.

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	03 H

Routine Results Request:

The status of the actuator test shall be checked using the request below:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest PR	31 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	03 H

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Positive Response:

On receiving valid RequestRoutineResults request, below response shall be sent.

ECU ⇒ Tester: RoutineControlPositiveResponse		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	03 H
5	Routine Status	xx H
6	Result Byte: Actuator 1	xx H
7	Result Byte: Actuator 2	xx H
8	Result Byte: Actuator 3	xx H
9	Result Byte: Actuator 4	xx H

Definition of Routine Status:

Value	Status
0x01	Routine is running
0x02	Routine has successfully finished
0x04	Routine has been stopped
0x08	Routine has been interrupted due to an error.

Note:

- A DHP error is set if the actuation limit exceeds the permitted level mentioned below:

Component	Maximum Actuation Time (in seconds)
Inlet Valve - EV	60
Outlet Valve -AV	30
Pump motor	30

- The Valve/Pump time limitation up counter starts when there is an actual requested actuation value (i.e., greater than 0). Once the defined time limit is reached, the protection state shall be set to Valve/Pump failed and a failure information is also set accordingly till the protection state is set.
- It takes twice the time for the down counter as that of the up counter to release the protection state after a failure.
- For example, for an EV failure, it takes 60s to set the protection state and 120s to release it.
- Any actuation request during the protection state set time shall be suppressed and hence shall not be processed.

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14.4 Wheel Speed Sensor Test

Purpose:

This test shall be used to check connection of the sensors to the corresponding wheel and for checking signal quality.

Entry and Exit conditions:

Entry condition:

The routine shall not start if any of the below specified conditions are met:

1. Failure pertaining to wheel speed sensors.
2. Other routine is running.

Exit condition:

The ongoing routine comes to a stop if the below conditions are encountered:

1. If a WSS related failure is set

Process:

The Wheel Speed Sensor test routine checks all the wheel speeds for a specified time and provides the maximum and minimum wheel speed in the given interval.
The test duration is chosen to be equivalent to one complete wheel revolution.

Start Routine Request:

The request of WSS routine is shown here:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	04 H
5	Execution Time (HB)	xx H
6	Execution Time (LB)	xx H

Execution Time: 1 LSB = 10 ms

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Positive Response:

On a valid request and no failures (mentioned above), the ECU shall send the following positive response:

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	04 H

Stop Routine Request:

The request to stop WSS routine is shown here:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	04 H

Positive Response:

On receiving valid stop routine request, below response shall be sent:

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	02 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	04 H

Request Routine Results:

WSS routine status shall be checked using the below request:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	04 H

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Positive Response:

On receiving a valid RequestRoutineResults request, the status of the WSS routine shall be received as shown below:

ECU ⇒ Tester: Routine Control Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	03 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	04 H
5	Routine Status	xx H
6	Max. Wheel Speed Front (HB)	xx H
7	Max. Wheel Speed Front (LB)	xx H
8	Min. Wheel Speed Front (HB)	xx H
9	Min. Wheel Speed Front (LB)	xx H
10	Max. Wheel Speed Rear (HB)	xx H
11	Max. Wheel Speed Rear (LB)	xx H
12	Min. Wheel Speed Rear (HB)	xx H
13	Min. Wheel Speed Rear (LB)	xx H

Definition of Routine Status:

Value	Status
0x01	Routine is running
0x02	Routine has successfully finished
0x04	Routine has been stopped
0x08	Routine has been interrupted due to an error.

Note:

- Resolution of speed v = 0.05625 Km/h.
- If the routine is stopped by giving the stop routine command (31 02 B0 04), the request result response shall contain the speeds calculated until the time the routine was stopped.
- If the routine has been interrupted due to an error (say, WSS failure), then the request results shall respond with FFFFH for the speeds to indicate that the routine was stopped due to a failure in the system.

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14.5 ABS SILA Lamp Blink Test

Purpose:

This test is for Original Equipment Manufacturer (OEM) to check the ABS SILA at their End Of Line (EOL). This test routine blinks the SILA based on a specific pattern.

Entry condition:

If any of the following conditions are met, the routine shall not start, and ECU shall send the ConditionsNotCorrect NRC (22H):

1. Any current failure
2. Other routine is running.

Start Routine Request:

The ABS Sila Blink Test shall be requested using the below command:

Tester ⇒ ECU: Routine Control Request		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	RoutineControl Type	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	05 H

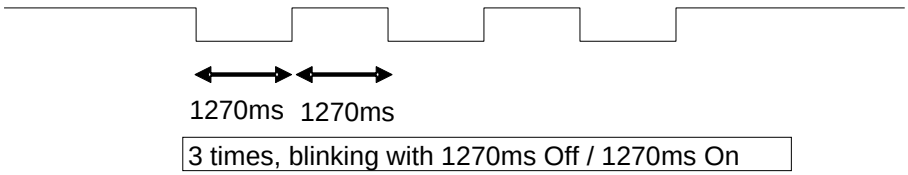
Positive Response

On a valid request and no failures (mentioned above), the ECU shall send the following Positive response:

ECU ⇒ Tester: Routine Control Positive Response		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineControlType	01 H
3	Routine Identifier HB	B0 H
4	Routine Identifier LB	05 H

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The SILA blinking pattern is as shown below. Note that this pattern starts with “ON” because this routine is done in the extended session where the SILA is already turned ON.



Note:

- Stop and Result routines are not applicable for ABS Sila Test.

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14.6 Sensor Mounting Validation Test

Purpose:

This routine is done to validate the mounting of the IMU sensor onto the bike. A comparison between the expected values of lean and pitch angles and the learned values are done to check if the values are within a tolerance of 2 and 4 degrees for pitch and lean respectively.

The routine shall be executed only after the PDMs corresponding to EOL mode, EOL Expected lean Angle and EOL expected Pitch Angle are initialised using the WriteDataByIdentifier Service (Refer section 13.7, 13.8 and 13.9).

Entry Condition:

The routine shall not start if any of the below specified conditions are met:

1. Any current failure relating to IMU sensor.
2. Other routine is running.
3. If EOL Mode is not initialised to 1

Start Routine Request

The Sensor mounting test shall be requested using the below service:

Tester ⇒ ECU: StartRoutineByLocalIdentifierRequest		
Byte	Content	Value
1	RoutineControlRequest SID	31 H
2	routineControlType	01 H
3	RoutineLocal Identifier	B0 H
4	EOLRoutineIdentifier	06 H
5	EOL Phase	01 H

Process:

The above request is to execute the “EOL phase 1”. In this process, ECU evaluates the lean angle and pitch angle for 10 samples and the average shall be compared against a predefined value (expected lean and pitch angle). If the calculated lean angle and pitch angle is not within range of the expected lean angle and pitch, then NRC22 shall be sent and EOL shall be locked.

On completion of Phase 1, the system exits from EOL with below positive response.

ECU ⇒ Tester: StartRoutineByLocalIdentifierPositiveResponse		
Byte	Content	Value
1	RoutineControlRequest PR	71 H
2	RoutineLocalIdentifier	B0 H
3	EOLRoutineIdentifier	06 H

Note:

- Stop and Result routines are not applicable for Sensor Mounting Validation Test.

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Negative response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: RoutineControlNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	Routine control Request	31 H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x12	Sub-functionNotSupported	This NRC shall be sent if the requested sub-function is either generally not supported or is not supported for the requested Routine Identifier.
0x22	ConditionsNotCorrect	This NRC shall be returned if the entry criteria for the request Routine Control are not met.
0x31	RequestOutOfRange	Invalid routine Identifier and RoutineControlOptionRecord
0x24	RequestSequenceError	1. The routine is not currently active when the 'stop Routine' sub-function is received. 2. Routine results are not available when the 'RequestRoutineResults' sub-function is received.
0x7F	ServiceNotSupportedInActiveSession	This NRC indicates that the requested action shall not be taken because the ECU does not support the requested service in the session currently active.
0x33	SecurityAccessDenied	This NRC shall be sent if the dataIdentifier, which reference a specific address, is secured and the ECU is not in an unlocked state.

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15 Communication Control

The tester shall use this service to control communication on the CAN bus. The purpose of communication control is to switch on/off the transmission and/or the reception of certain messages of ECU(s).

CommunicationControl Request:

Tester ⇒ ECU: CommunicationControlRequest		
Byte	Content	Value
1	CommunicationControlRequest SID	28 H
2	SubFunction	xx H
3	CommunicationType	xx H

Definition of SubFunction:

Sl no	SubFunction Value	SubFunction	Description
1	00 H	EnableRxAndTx	This value indicates that the reception and transmission of messages shall be enabled for the specified communicationType. SuppressPosRspMsgIndicationBit=FALSE
2	01 H	EnableRxAndDisableTx	This value indicates that the reception of messages shall be enabled and the transmission shall be disabled for the specified communicationType. SuppressPosRspMsgIndicationBit=FALSE
3	80 H	EnableRxAndTx	This value indicates that the reception and transmission of messages shall be enabled for the specified communicationType. SuppressPosRspMsgIndicationBit=TRUE
4	81 H	EnableRxAndDisableTx	This value indicates that the reception of messages shall be enabled and the transmission shall be disabled for the specified communicationType. SuppressPosRspMsgIndicationBit=TRUE

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Definition of CommunicationType:

Sl no.	CommunicationType Value	CommunicationType	Description
1	0x00 H	ISOSAEReserved	-
2	0x01 H	NormalCommunicationMessages	This value references all application-related communication (interapplication signal exchange between multiple in-vehicle ECU s).
3	0x02 H	NetworkManagementCommunicationMessages	This value references all network management related communication.
4	0x03 H	NetworkManagementCommunicationMessages and NormalCommunicationMessages	This value references all network management and application-related communication.

Note:

- For the current project implementation, there is no network management supported, hence communication is enabled/disabled for communication type 0x01H and 0x03H only.
- Session change from extended to default session shall enable reception and transmission of all messages.

Positive Response:

On a valid request, the ECU shall respond with following positive responses:

ECU ⇒ Tester: CommunicationControlPositiveResponse		
Byte	Content	Value
1	CommunicationControl Request PR	68 H
2	SubFunction	xx H

Note:

- If suppressPosRspMsgIndicationBit=TRUE, no response shall be sent.

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Negative response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: RoutineControlNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	Communication Control	28 H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x12	Sub-functionNotSupported	This NRC shall be sent if the sub-function parameter is not supported.
0x31	RequestOutOfRange	This NRC shall be returned if CommunicationType is not in requested range.
0x7F	ServiceNotSupportedInActiveSession	This NRC indicates that the requested action shall not be taken because the ECU does not support the requested service in the session currently active. Occurs when service is requested in default session.

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16 Control DTC Setting

The tester shall deactivate and activate the storage of DTCs using this service. This service shall be used by a tester to stop or resume the updating of DTC status bits in the ECU(s).

ControlDTCSetting Request:

Tester ⇒ ECU: ControlDTCSettingRequest		
Byte	Contents	Value
1	ControlDTCSetting	85 H
2	SubFunction	xx H

Definition of SubFunction:

Sl no	SubFunction Value	Description
1	01 H	DTC storage is enabled; suppressPosRspMsgIndicationBit=FALSE
2	02 H	DTC storage is disabled; suppressPosRspMsgIndicationBit=FALSE
3	81 H	DTC storage is enabled; suppressPosRspMsgIndicationBit=TRUE
4	82 H	DTC storage is disabled; suppressPosRspMsgIndicationBit=TRUE

Positive Response:

On valid request, the ECU shall respond with following positive response:

ECU ⇒ Tester: ControlDTCSettingResponse		
Byte	Content	Value
1	ControlDTCSetting PR	C5 H
2	SubFunction	xx H

Note:

- If suppressPosRspMsgIndicationBit=TRUE, no response shall be sent.

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Negative response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ Tester: ControlDTCNegativeResponse		
Byte	Contents	Value
1	Negative Response	7F H
2	Control DTC setting	85 H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLengthOrInvalidFormat	This NRC shall be sent if the length of the message is wrong.
0x12	Sub-functionNotSupported	This NRC shall be sent if the sub-function parameter is not supported.
0x7F	ServiceNotSupportedInActiveSession	This NRC indicates that the requested action shall not be taken because the ECU does not support the requested service in the session currently active. Occurs when service is requested in default session.

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17 Tester Present

The “Tester Present” service shall be used to maintain a diagnosis session (other than the default Session), when no other service is being executed. This service is used to indicate to an ECU (or ECUs) that a tester is still connected to the vehicle and that certain diagnostic services and/or communication that have been previously activated are to remain active.

The ECU shall reset S3 Server (see “Communication Parameters” section) to monitor S3 max again after sending the positive response.

Tester Present Request:

TESTER ⇒ ECU: TesterPresentRequest		
Byte	Contents	Value
1	TesterPresentRequest	3E H
2	ZeroSubFunction	xx H

Definition of ZeroSubFunction :

suppressPosRspMsgIndicationBit (7th Bit of ZeroSubFunction)	Value	Description
FALSE	00 H	Positive response is obtained
TRUE	80 H	Positive response is suppressed

Positive Response:

On valid request, the ECU shall respond with the following positive response:

TESTER ⇒ ECU: TesterPresentPositiveResponse		
Byte	Contents	Value
1	Tester PresentPR	7E H
2	ZeroSubFunction	00 H

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Negative Response

Negative response code shall be sent in the format as indicated below:

ECU ⇒ TESTER: Tester Present Negative Response		
Byte	Contents	Value
1	Negative Response	7F H
2	Tester Present	3E H
3	Response Code	xx H

Response Code	Description	Reason
0x13	IncorrectMessageLength Or Invalid Format	This NRC shall be sent if the length of the message is wrong.
0x12	Sub-functionNotSupported	This NRC shall be sent if the sub-function parameter is not supported.

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18 Appendix A: List of Actuators

Actuator No. (Hex)	Content	*	**	Actuation Value
0000	No activation			
0022	Pump-Motor	d	i/o	0xFFFF: ON 0x0000: OFF
0030	EV front	d	i/o	0xFFFF: ON 0x0000: OFF
0032	AV front	d	i/o	0xFFFF: ON 0x0000: OFF
0038	EV rear	d	i/o	0xFFFF: ON 0x0000: OFF
003A	AV rear	d	i/o	0xFFFF: ON 0x0000: OFF
0064	Disable VR-shut down	d	o	0xFFFF: Disable shut down. 0x0000: Enable shut down.

Note:

- *: a = Analog Value; d = Digital Value
- **: i = input signal; o = output signal; i/o = input- and output-signal

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19 Appendix B: List of Abbreviations

- ABS - Anti-lock Braking System
- MTC - Motorcycle Traction Control
- DTC(Feature Control) – Drag Torque Control
- CC - Cruise Control
- LC - Launch Control
- DTC - Diagnostic Trouble Code
- UDS - Unified Diagnostic Services
- CAN - Controller Area Network
- ECU - Electronic Control Unit
- ROM - Read Only Memory
- EOL - End of Line
- RDBI - Read Data by Identifier
- WDBI - Write Data by Identifier
- HB - High Byte
- MB - Middle Byte
- LB - Low Byte
- AV - Outlet Valve
- EV - Inlet Valve
- EEPROM - Electronically Erasable Programmable Read Only Memory
- PR - Positive Response
- NRC - Negative Response Code
- DID - Data Identifier
- LSB - Least Significant Byte
- SID - Service Identifier
- DHP - Diagnostic Hydraulic Protection
- WSS - Wheel Speed Sensor

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